A Small Experiment, Clearly Reported

Texite Demo Project

Abstract

We compare three settings for a small prediction task. This note keeps the question, metric, results, and limitations together. All values are synthetic and illustrate a writing workflow, not a scientific finding.

1 Question and method

Does using a larger training set improve prediction quality? We compare the same model using 100, 300, and 500 training examples. The evaluation metric is the mean absolute error:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{1}$$

A lower value is better. Table 1 records the comparison.

2 Results and limitations

The larger training set gives a lower error in this example.

Table 1: Illustrative results. All numbers are synthetic.

Setting	Training examples	MAE
Small	100	0.42
Medium	300	0.31
Large	500	0.27

Before drawing a conclusion, repeat the experiment with several random seeds and report the variation. The figure on the next page provides a compact view of the same values.

3 A figure that stays connected

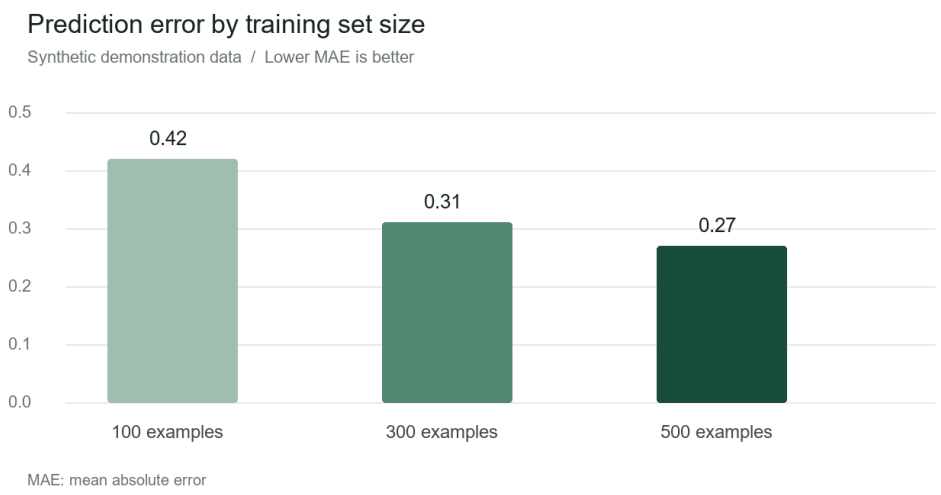


Figure 1: The same synthetic values as Table 1.

Keep the data, figure, and text consistent when revising a result. Equation 1 defines the metric; Section 1 describes the comparison. Cross-references update during compilation. For background on document structure and references, see Lamport [1].

3.1 Before sharing

- Check that all citations and figure references resolve.
- Describe limitations alongside the result.
- Create a project backup before a substantial revision.

References

[1] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2 edition, 1994.